

RESEARCH ARTICLE

Animal Colonization of Restored Peatlands: Inoculation of Plant Material as a Source of Insects

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Abstract

Ecological restoration of mined peatlands in North America involves active reintroduction of bog plant species. Animals are not actively reintroduced, thus the re-establishment of peatland fauna must occur either by inoculation along with introduced plant material or by dispersal. We examined the extent to which insects are reintroduced to restored sites with plant material by rearing insects from shredded vegetation collected in three donor sites. We assessed differences in abundance, diversity, and composition of taxonomic and trophic groups among seasons and sites. Abundance and species richness did not differ by season, but species assemblages did. The three sites were significantly different in abundance, but not in species richness and assemblages. Few insects emerged

from the vegetation, suggesting that shredded plant material may not be the primary source of insect colonists. Insects likely recolonize by active or passive dispersal from the surrounding area. The species pool was similar among donor sites; consequently a mined site could be inoculated with vegetation from another peatland in the same region and this would not affect the insect assemblages at the initial stage of establishment. Diapause may be a major factor for emergence success among seasons of collection. Knowledge of how restoration techniques influence establishment of insect communities will help predict longer-term outcomes of restoration on biotic communities in peatlands.

Key words: biodiversity, emergence, recolonization, restoration, species assemblages.

Introduction

Ecological restoration of mined ombrotrophic peatlands in North America consists of facilitating the recovery of a peat-accumulating ecosystem (Rocheftort & Lode 2006). Because abandoned vacuum-mined fields are not naturally recolonized by *Sphagnum* mosses (Rocheftort 2001; Poulin et al. 2005), active reintroduction of peatland species is necessary (Rocheftort et al. 2003). Plants are recruited from material collected in donor sites (natural peatlands or natural remnants in mined areas), shredded and inoculated in restored sites. In contrast, animals are not actively reintroduced; thus re-establishment of peatland fauna must occur by inoculation with introduced plant material or by active and/or passive dispersal post-restoration.

Emergence patterns of insects from shredded vegetation after anthropogenic disturbance are likely to differ from patterns in undisturbed sites. Aspects of the restoration process (depth of collection, shredding and stockpiling of vegetation) could have a direct impact by killing the insects or altering requirements for breaking dormancy.

Anthropogenic disturbances such as agricultural practices and chemical contamination disrupt the emergence of invertebrate egg banks in wetlands (Hathaway et al. 1996; Barry & Logan 1998; Gleason et al. 2003; Barry & Davies 2004). There could also be an indirect impact on the survival of organisms because of unsuitable environmental conditions at the restored site (Gleason et al. 2003; Nielson et al. 2003; Angeler & García 2005).

Community assembly and structure at a site are influenced by availability of colonist sources and post-emergence biotic interactions (Angeler & García 2005). After restoration, the established community may change noticeably through competition and predation as biotic and abiotic characteristics recover. If recovery of insects results from reintroduction as well as from refugial populations and/or dispersal from surrounding sources, reintroduced insects could be an important colonist source for small species with low dispersal capabilities (Gleason et al. 2004). Moreover, these early species may play important functional roles for plant establishment, improving substrate quality and filling niches not occupied by species established on unrestored sites.

The objectives of this study were to determine the extent to which insects are actively reintroduced to restoration sites with plant material collected on donor sites and to assess differences in abundance, diversity and composition of insect species and trophic groups among seasons and sites.

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Methods

Collection of Plant Materials

Three peatlands in the St. Lawrence river estuary, Quebec, were chosen because they have served or could serve as donor sites for restoration of abandoned mined peatlands: the natural section on the north side of St-Charles bog (SC) (46°45'53.9"N, 70°59'58.5"W); the remnant natural section of Verbois bog (VE) (47°50'08.5"N, 69°27'22.9"W) and in St-Alexandre bog (SA) (47°44'24.7"N, 69°36'43.8"W). The sites were dominated by *Sphagnum* mosses from the groups *Acutifolia* and *Cuspidata*, ericaceous shrubs, trees, other mosses such as *Polytrichum* and lichens. Because peat extraction for restoration is usually done in early summer or in fall, plant material was collected on 11 June 2008 (J) and 23 September 2008 (S), to allow comparison of the effect of season on insect assemblages collected. Plant material was collected from one transect in each field, established where plant species favorable for restoration (suitable *Sphagnum* mosses and low shrubs) were dominant (Quinty & Rochefort 2003). The top 10 cm of plant cover (Quinty & Rochefort 2003) was collected in four quadrats (25 × 25 cm) every 8 m along the transect. Each sample was cut into 1–3 cm pieces using hedge-trimming shears to mimic disturbance caused by the restoration techniques, as the surface vegetation is shredded by a tractor prior to collection (Quinty & Rochefort 2003).

Insect Collection and Processing

Plant material was measured for volume (mL), spread evenly in trays (26 × 51 × 5 cm) and covered with a mesh emergence cage (black sides and white top) over a wood frame (26 × 51 cm, 36 cm high). Emergence cages were placed at 20 ± 2°C with 15 hours of day length to initiate development after diapause (McLeod et al. 1985; Hinks & Doane 1988). Emergence from June samples was monitored for 14 weeks. September samples were left at 20 ± 2°C for at least 5 weeks then stored at 4°C for 12 weeks, to induce diapause. Samples were then left at 20 ± 2°C for 14 weeks for emergence.

Emerging insects were collected in a 5-cm diameter glass jar filled with salt water and propylene glycol attached to the top corner of the cage. Jars were sampled monthly and specimens were preserved in 70% ethanol. Small insects were identified in ethanol while larger ones were pinned. Specimens were deposited in the Lyman Entomological Museum (McGill University, Ste-Anne-de-Bellevue, QC, Canada). All specimens except Microlepidoptera and Collembola, were identified to named species if published keys or taxonomic expertise were available, or to numbered morphospecies. Collembola were excluded because the collecting method was inappropriate for quantitative assessment of their diversity. Microlepidoptera were excluded because of the difficulty in handling and identifying specimens in alcohol. Female specimens that could not be separated into morphospecies were excluded from analyses. Species were classified in one of seven trophic groups, based primarily on larval feeding habits: saprophagous (SA), predator (PR), parasite (PA), parasitoid (PO), phytophagous

(PH), omnivore (OM) and unknown (?), and separated between trophic generalist (G) and specialist (S) (Table S1, Supporting Information).

Statistical Analysis

Two-way analyses of variance (ANOVA) was used to assess the effect of site and season on the total insect abundance and species richness. Analyses were performed on four emergence trap replicates per season per peatland site. A series of analyses were performed with one-way ANOVA to examine the differences in abundance and species richness of individual orders, and between generalists and specialists among the two seasons and sites. Two-way ANOVA without replication was performed on trophic groups for each site for the difference in abundance and species richness between groups and between seasons. Prior to analyses, data were log ($n + 1$) transformed to stabilize variance. A simple linear regression using the least squares method was used to assess the relationship of insect abundance and richness to volume of peat (mL). These analyses were performed with Excel 2007 Analysis ToolPak (Microsoft Corporation, Redmond, WA, U.S.A.).

Data from the four replicates in each peatland were pooled for analyses of diversity. The number of individuals, observed species richness, number of species in each trophic group, and Simpson's diversity index were calculated using EstimateS version 8.2.0 (Colwell 2009). Rarefaction estimates standardized to 33 individuals were calculated based on 1000 iterations with species richness as a diversity index with ECOSIM version 7.0 (Gotelli & Entsminger 2001).

The species assemblages in the peat samples at each site per season were compared using non-metric multidimensional scaling (NMDS) and multi-response permutation procedure (MRPP). The matrices were log ($n + 1$) transformed prior to analyses. For NMDS, a six-dimensional solution was first run with detrended correspondence analysis (DCA) ordination that was previously run on the data as the starting coordinates. A final NMDS with 400 iterations and 200 runs with real data was then run with the DCA ordination as starting coordinates and the number of dimensions giving the lowest final stress. A Monte Carlo test with 500 randomized runs was executed to evaluate if NMDS extracted stronger axes than expected by chance ($p \leq 0.05$). For MRPP, a Sorenson distance metric was applied, to test for pair-wise differences in species composition for seasons and sites. Analyses were performed using PC-ORD version 5.31 (McCune & Mefford 2006).

Results

A total of 573 individuals were collected, comprising 112 species from 31 families, of which 70 species were singletons. Diptera and Hymenoptera were the most species rich orders collected, and a higher proportion of Diptera was collected in June and September, followed by Hymenoptera (Fig. 1). The most abundant species in June were *Corynoptera ?perfecta* (Petty) (Diptera: Sciaridae), *Tapinoma sessile* Say

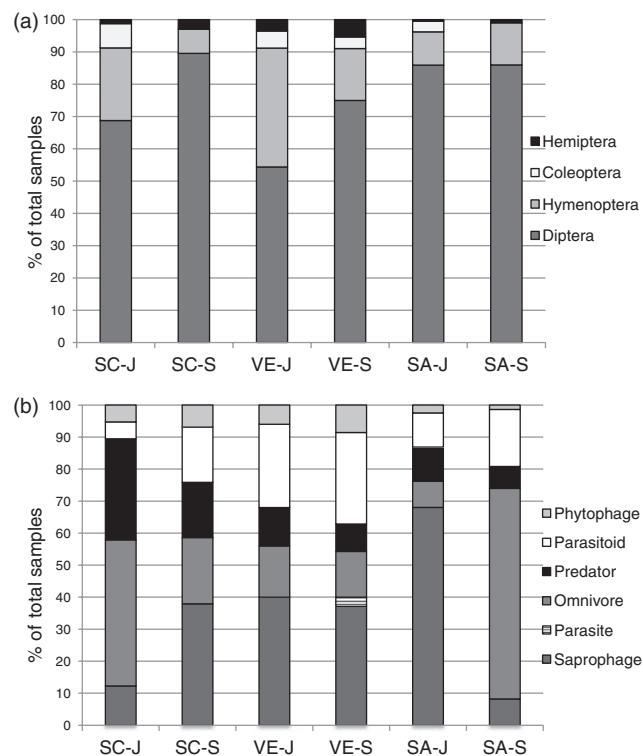


Figure 1. Relative proportion of (a) individual orders and (b) trophic groups for St-Charles (SC), Verbois (VE), and St-Alexandre (SA) peatlands in June (J) and September (S) samplings.

(Hymenoptera: Formicidae), and *Cantharis* sp.1 (Coleoptera: Cantharidae), while the most abundant in September were all Diptera: Orthocladiinae sp.7 (Chironomidae), *Corynoptera* sp. (Sciaridae) and *Bradysia* (*bicolor* group) sp.1 (Sciaridae). In all sites, generalists were more abundant than specialists, although the number of species was similar. Relative abundance and observed species richness were higher in June than September for all sites (Table 1). However, season did not affect abundance ($p = 0.23$) or species richness ($p = 0.065$). The three sites were significantly different in abundance ($p = 0.045$), although not in species richness ($p = 0.22$). Abundance and species richness of individual orders, trophic groups, and generalists versus specialists were not significantly different between June and September and between the three sites ($p > 0.05$). Abundance and species richness among trophic groups within each site was significantly different (Fig. 1). No significant treatment effect was found on abundance or species richness for the volume of peat collected ($p > 0.05$). Rarefaction analysis showed that diversity did not differ among June and September for all three sites. The Simpson's diversity index was lower in June for St-Charles and in September for Verbois and St-Alexandre (Table 1). The low estimated diversity and Simpson's index in St-Alexandre was due to the dominance of *Corynoptera* ?*perfecta* in June and Orthocladiinae sp.7 in September. The abundance and species richness among the trophic groups within each site was significantly different (Fig. 1).

Table 1. Abundance, species richness, rarefaction estimates (standardized at 33 individuals) (S_{est}) and Simpson's diversity index for St-Charles (SC), Verbois (VE), and St-Alexandre (SA) peatlands in June (J) and September (S) samplings.

Sample	Abundance	Richness	S_{est}	Simpson
SC-J	61	33	20 ± 1.96	13.26
SC-S	40	24	21 ± 1.18	19.02
VE-J	53	27	19 ± 1.69	18.62
VE-S	38	21	19 ± 1.05	14.96
SA-J	131	29	12 ± 1.9	3.15
SA-S	78	20	11 ± 1.72	2.95

There were significant differences in species assemblages. June and September collections were clearly separated in the one-dimensional NMDS ordination ($p = 0.012$; final stress = 43.4) and were significantly different in MRPP among the three sites ($p < 0.01$). Species assemblages in SC-S and VE-S, SC-S and SA-S, VE-J and SA-J, VE-S and SA-S did not differ significantly in Pair-wise MRPP (Table 2). June species assemblages from St-Charles were significantly different from those in Verbois and St-Alexandre, which did not significantly differ from each other. September species assemblages were similar for the three peatlands. Grégoire Taillefer and Wheeler (2012) identified 446 species of Brachycera (Diptera) in restored peatlands in eastern Canada. In this study, 15 Brachycera species emerged from the peat samples collected at the three donor sites. Three of these species were also identified by Grégoire Taillefer and Wheeler (2012) at the restored sites 7 years after restoration: *Coenosia* (*Limosia*) ?*bonita* Hockett (Muscidae), *Hybomitra* *typhus* (Whitney) (Tabanidae) and *Thaumatomyia* *pulla* (Adams) (Chloropidae). These species are predacious and *H. typhus* is known to predate in the substrate.

Discussion

The low insect emergence suggests that shredded vegetation used in peatland restoration may not be a primary source of insect colonists. Scouring of soil from drainage ditches for transplantation at restored sites reduces the density of macroinvertebrates (Brown et al. 1997) and even a 0.5 cm burial by erosion reduced aquatic invertebrate emergence by 99.7% (Gleason et al. 2003). Direct transfer of plant and insect propagules may, however, decrease the time needed

Table 2. p -Values for MRPP pair-wise comparisons among the two sampling periods for St-Charles (SC), Verbois (VE), and St-Alexandre (SA) peatlands in June (J) and September (S) samplings.

Sample	SC-S	VE-J	VE-S	SA-J	SA-S
SC-J	0.023	0.0097	0.009	0.013	0.01
SC-S		0.024	0.1	0.01	0.32
VE-J			0.04	0.7	0.036
VE-S				0.0067	0.13
SA-J					0.024

Significant differences ($p \leq 0.05$) in species composition between treatments are in bold.

for community convergence to a pre-disturbance level. Watts et al. (2008) found that beetle community recovery was most rapid when vegetation was directly transferred in mined bogs.

The early stages of restoration of newly created peatlands may be characterized by a limited pool of Brachycera (Diptera) species. Fifteen species of Brachycera were collected at the donor sites. Given the high abundance and diversity (446 species) of higher Diptera in restored sites in our previous study (Grégoire Taillefer & Wheeler 2012), and their role in ecosystem function, it seems that dispersal from the surrounding area may play a major role in the re-establishment of a peatland insect community, either passively on the wind, or actively by flying or walking. This is ecologically interesting, but also has conservation and management implications for peatland restoration. Because dispersal limitation is frequently cited in differences between natural and restored sites (Brown et al. 1997; Tschamtkke et al. 2002), distance to an undisturbed habitat should be considered. Watts and Didham (2006) found that more isolated sites supported a decreased species richness and abundance of colonizing invertebrates. Connectivity to natural patches is also of relevance (Scott et al. 2001) as some species require corridors to colonize new habitats (Collinge 2000). If such natural sites are not close to restored sites, then other methods such as active reintroduction may be required to aid in the re-establishment of the peatland-associated insect community.

The species pool in the samples was similar among sites, based on abundance, species richness, and composition; only St-Charles in June had a different assemblage. Trophic groups, individual orders and generalists versus specialists species richness and abundance exhibited similar patterns to species community structure. Peatland insect communities are specialized to a humid environment, an acidic pH and nutrient deficiency with characteristic vegetation (Payette 2001); the species composition does not vary regionally. A mined site to be restored could be inoculated with vegetation from another peatland in the same region if there is no natural remnant of this peatland and this would not affect the insect species assemblages at the initial stage of establishment. Larvae of most of the species collected are specific to moist habitats (Stehr 1987–1991) or are parasites of larvae in moist habitats.

Some species may depend on translocation of peat material to colonize the new habitat because of limited dispersal ability due to small body size (e.g. Sciaridae, Chironomidae, and Cecidomyiidae). Those groups may be important in post-restoration succession through a typical peatland community as species at the top of the food web and specialists, such as predators and parasitoids, are usually more susceptible to isolation, habitat quality, and size (Tschamtkke et al. 2002). Chironomids are an important food source in aquatic communities and may result in faster recolonization of restored sites by predatory invertebrates or birds (Tokeshi 1995). Brady et al. (2002) also found that inoculation of a wetland site with plant material collected from a natural wetland assisted a few invertebrate taxa, mainly gastropods and invertebrates without aerial dispersal abilities.

Differences resulting from collection date suggest that diapause may be a major factor in emergence success. If vegetation is collected and spread on the bog in September, mortality of diapausing stages is likely to be higher because of exposure on the surface. Freeze and desiccation tolerance of species would influence winter survival. Emergence and establishment might be more likely if inoculation was done in June and this possibility should be explored further. However, total, individual order and trophic abundance and species richness were similar for both seasons, which indicate that the rate of emergence is similar for both seasons. The difference in species assemblages is another issue. The September collections probably contained more individuals in diapause than the June collections, which probably contained more active adults and individuals ready to terminate diapause. Emergence from September was spread out over 19 weeks unlike June collections where the highest proportion of insects was collected in the first month (Table S1). Most of the species in September collections (e.g. Sciaridae, Chironomidae, and Cecidomyiidae) may overwinter in the peat and be able to survive penetration by external ice (Kostal & Havelka 2000) and low temperature (Tanno 1977; Baust & Edwards 1979). However, species in June collections may feed in the peat, but overwinter elsewhere (e.g. *Tapinoma sessile*). This may explain the difference in species composition found between the two collection seasons.

Knowledge of how restoration techniques influence the establishment of insect communities will help predict longer-term outcomes of restoration on biotic communities in peatlands. Because these restored ecosystems are also of conservation concern, the recovery of a typical species composition should remain the primary goal and this is usually achieved through patterns of colonization and extinction. It is likely that monitoring emergence in the field immediately following translocation of peat material to restored sites and comparison of emergence to donor sites may give more precise information on the insect population present at the first stage of restoration. Environmental and habitat variables, such as percentage of *Sphagnum* mosses, plant species richness and litter biomass should also be tested for variation in insect community composition and abundance in peat transplant.

Implications for Practice

- Translocated vegetation may not be the primary source of insect colonists in restored peatlands.
- Proximity of natural peatland patches to sites to be restored is important as a potential source of insect colonists for community recovery.
- If adjacent natural peatland patches are absent, active reintroduction of insects to restored sites may be required for a successional pathway leading to a typical peatland assemblage.
- Vegetation should be collected and spread in early summer to maximize insect emergence success because of higher winter mortality.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Insect species and morphospecies collected from St-Charles (SC), Verbois (VE) and St-Alexandre (SA) peatlands in June (J) and September (S) samplings. Taxon names are followed by the feeding behavior (specialist [S] and generalist [G]), the trophic group (saprophage [SA], predator [PR], parasite [PA], parasitoid [PO], phytophage [PH], omnivore [OM], and unknown [?]) and the number of specimens collected in the four emergence cages pooled.